D2C FASHION CUSTOMER INTELLIGENCE

Decoding Customer Value

Is the business building genuine customer loyalty, or relying too heavily on continuous promotion — and what should management do next?

— Case Study by **Yash Pratap Singh**

EXECUTIVE ANSWER

Behavioral Loyalty Signals Are Concentrated — Not Uniform Across the Base

Repeat behavior, not current spend, separates strong customers from weak ones — so blanket promo removal is not supported.

6.5 → 36.6

Repeat-Purchase History range across six behavioral segments

$59.24 → $60.93

Current Observed Spend range — nearly flat by comparison

43.0%

of the base (1,677 of 3,900) carries Observed Promo Exposure

- Current promo exposure alone does not identify strong or weak customers; current promo users show materially different repeat-purchase histories.
- Current promo users are not one group: Higher-History Current Promo Users (31.4 avg. Repeat-Purchase History) look nothing like Lower-History Current Promo Users (16.2), despite near-identical spend.

Recommendation: replace blanket discounting with a controlled promotional-sunset pilot on the best-evidenced segment, protect lower-history promo users, and build longitudinal measurement in parallel.

Before the Answer: What This Data Can and Can't Prove

This is a customer-level snapshot without transaction history, so loyalty is approximated, not measured directly.

CORE BUSINESS QUESTION

Is the company building genuine customer loyalty, or relying too heavily on continuous promotion — and what should management do next?

What the data CANNOT show

- **No transaction timestamps** — no actual retention, customer departure, or causal promotional effect
- **Stated purchase frequency** is self-reported, not transaction-derived
- **One observed purchase** and one category per customer; state-level samples are small and cross-sectional
- **No acquisition cost,** market share, time trend, discount depth, or margin data

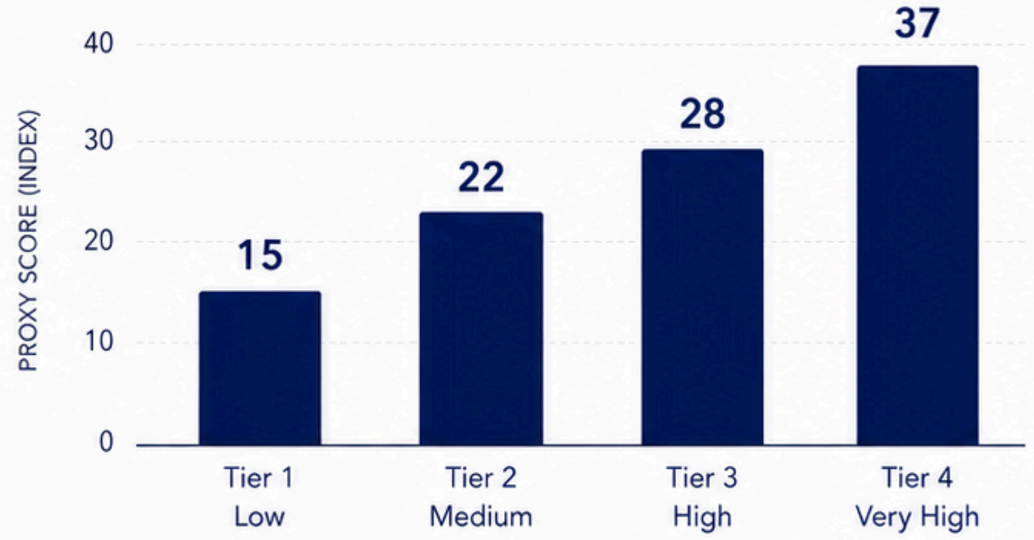
What the data CAN show

- **Current Observed Spend**, Repeat-Purchase History, and stated purchase frequency at a single point in time
- **Observed Promo Exposure** and subscription status as of the snapshot
- **Satisfaction,** category, geography, and demographics for context and segmentation
- **Behavioral Loyalty Proxy** and **Commercial Value Proxy** — analytical rankings built from the above

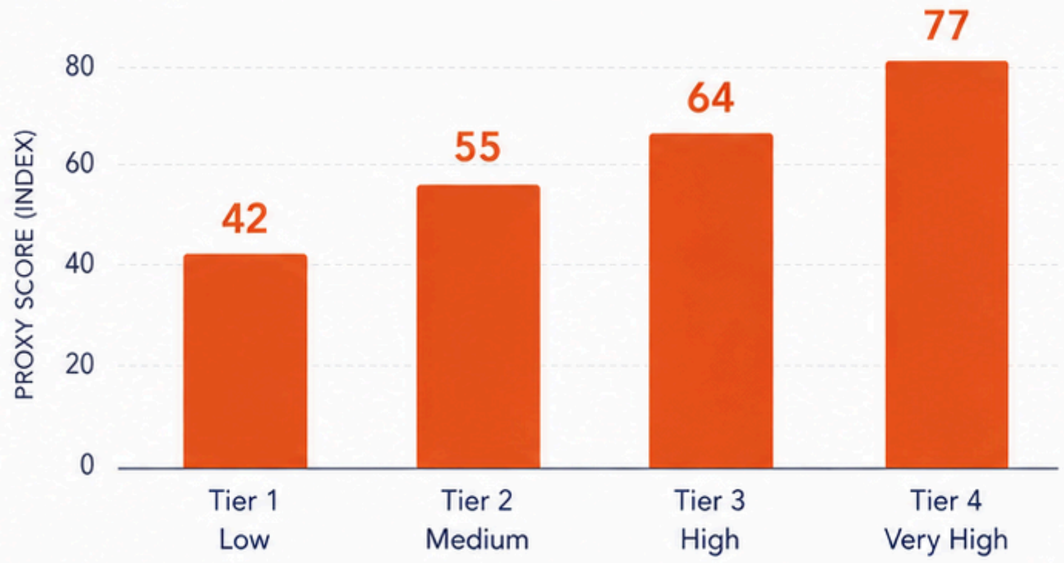
Two Proxies Answer Two Different Questions

The Behavioral Loyalty Proxy tracks engagement depth; the Commercial Value Proxy tracks current commercial contribution — and they relate to Observed Promo Exposure in opposite ways.

BEHAVIORAL LOYALTY PROXY TIERS



COMMERCIAL VALUE PROXY TIERS



Definition A ranks behavioral engagement with current promo exposure included as one component. Promo separation across its tiers is therefore partly mechanical and should not be treated as independent validation. **Definition B** provides a complementary commercial-value lens.

A Six-Tier Pyramid Ordered by Repeat-Purchase History

The two current-promo segments sit at very different pyramid levels — Observed Promo Exposure alone does not define the tier.

1	High Behavioral Loyalty		Highest Repeat-Purchase History tier (avg. 36.6) — minimal Observed Promo Exposure
2	Higher-History Current Promo Users	USERS	497 customers (12.7% of base) · 31.4 avg. history · 100% Observed Promo Exposure
3	Upper-Mid Behavioral Loyalty		Intermediate repeat depth, low current promo exposure
4	Lower-History Current Promo Users	USERS	16.2 avg. history · 100% Observed Promo Exposure
5	Lower Behavioral Loyalty — No Current Promo		Below-average repeat depth, not currently promo-exposed
6	Lowest Behavioral Loyalty — No Current Promo		Lowest tier — 6.5 avg. history, no current promo

05 | KEY DRIVER

Basket Size Doesn't Separate Customers. Repeat Behavior Does.

Current Observed Spend moves by roughly 3% across segments; Repeat-Purchase History moves more than 5x.

$59.24 → $60.93

Current Observed Spend range — all six segments

6.5 → 36.6

Repeat-Purchase History range — all six segments

Segment	Avg. Repeat-Purchase History	Avg. Current Observed Spend
High Behavioral Loyalty	36.6	within $59.24–$60.93 range
Higher-History Current Promo Users	31.4	$59.44
Lower-History Current Promo Users	16.2	$59.24
Lowest Behavioral Loyalty — No Current Promo	6.5	within $59.24–$60.93 range

06 | HETEROGENEITY

Same Current Promo Exposure, Very Different Customers

Among customers with identical 100% Observed Promo Exposure, Repeat-Purchase History nearly doubles between the two segments.

Metric	Higher-History Current Promo Users	Lower-History Current Promo Users
Observed Promo Exposure	100%	100%
Avg. Repeat-Purchase History	31.4	16.2
Stated Frequency Score	2.70	1.68
Satisfaction	3.81	3.54
Current Observed Spend	$59.44	$59.24

Treatment strategy must segment within current promo users — Observed Promo Exposure alone is not sufficient for decisions.

Geography Points to Test Markets — Not Proven Growth

State-level differences in spend and promo exposure exist, but with no acquisition cost, market share, or time-trend data, they identify candidate test cells only.

The geographic view plots Average Current Spend against Promo Exposure % by state.

Priority test markets sit above-average spend and below-average promo exposure — a targeting signal, not a growth forecast.

Each state is a small, cross-sectional sample; the data contains no acquisition cost, market share, or time trend.

Recommended use: build matched test/control cells by state for the promotional-sunset and messaging experiments — not for investment sizing.

Matched Test / Control Cells

Select states with adequate sample size and comparable baseline behavior before assigning treatment.

Category Differences Are Modest — Use for Context, Not Strategy

Repeat-Purchase History spans only 25.0–25.7 and Current Observed Spend only $57.17–$60.26 across categories.

25.0 – 25.7

Avg. Repeat-Purchase History range across categories

$57.17 – $60.26

Avg. Current Observed Spend range across categories

Purchase-history mix (low / mid / high history) is broadly similar across Accessories, Clothing, Footwear, and Outerwear.

Only one purchase category is observed per customer — the data cannot show cross-category behavior.

Recommended use: balance test cells and tailor merchandising or message relevance — not a basis for broad strategic claims.

A Three-Phase, Controlled Sunset Pilot

Test the sunset on the best-evidenced population first — Higher-History Current Promo Users — with a holdout at every phase.

1 Phase 1 — Pilot

- Randomize eligible Higher-History Current Promo Users (497 customers, 12.7% of base) into treatment (discount suppressed/reduced) vs. control (existing promotion retained).

Primary KPI: repeat-purchase incidence in a predefined observation window.

2 Phase 2 — Expand

- Expand only if the primary KPI is **non-inferior** to control.
- Monitor guardrails: satisfaction, refunds/returns, cancellations, support contacts, and gross margin once available.

3 Phase 3 — Scale

- Broader rollout by validated cell only, with a persistent holdout and periodic revalidation across geography and category subgroups.

Excluded from Phase 1: Lower-History Current Promo Users (16.2 avg. history) are a separate, lower-history population and require their own evidence before any change.

Stop rule: pause and restore the existing treatment if the non-inferiority bound or any guardrail threshold is breached.

Define the ICP by Behavior First, Demographics Last

Observed Promo Exposure and subscription status assign treatment — they are never mandatory ICP gates.

1	**Repeat-Purchase History**	Primary signal. The existing high-history flag begins at 38 purchases — the dataset's 75th-percentile threshold.
2	**Stated frequency + satisfaction**	Supporting Engagement Proxy signals; satisfaction rating of 4.0 or higher where available.
3	**Current Observed Spend / Commercial Value Proxy**	Used to prioritize within behaviorally similar customers — not as the primary qualifier.
4	**Observed Promo Exposure & subscription status**	Assign treatment and communication strategy only. Every subscriber in the source is also promo-exposed, so the two are structurally confounded.
5	**Category, geography, age, gender, payment method**	Context, creative relevance, and experimental balance only — not core qualification criteria.

One-sentence ICP: *A customer with deep Repeat-Purchase History, frequent stated purchasing and strong satisfaction, prioritized with Current Observed Spend where useful and treated according to current promo and subscription context rather than defined by demographics or a mandatory promo/subscriber status.*

Measurement Before Marketing: A 90-Day Build

Days 0–30 build measurement and experiment readiness; days 31–90 run and scale only the treatments that prove out.

Days 0–30
Build readiness

- Capture order timestamps, discount depth, promotion type, margin inputs, acquisition source
- Define eligibility, randomization, control, guardrails, and reversal rules
- Establish an auditable baseline from the current segment assignment

OWNER: Data/Analytics, Engineering, Growth/CRM, Finance

Days 31–60
Run controlled tests

- Launch the promotional-sunset pilot on eligible Promo-Engaged Repeaters
- Run a matched geographic test using comparable baseline cells
- Establish the first longitudinal repeat-purchase baseline

OWNER: Growth/CRM, Analytics, Customer Experience

Days 61–90
Scale the evidence

- Scale only treatments that remain non-inferior and within guardrails
- Build cohort and promotion history; construct a margin-aware value model
- Refine the ICP using newly observed post-launch behavior

OWNER: Analytics, Finance, Growth/Marketing

Operating principle: ***no treatment scales because of a favorable snapshot association alone.***

What Would Make This Analysis Stronger

This snapshot cannot prove retention, churn, or causal promo effects — closing that gap is the highest-value next investment.

Current limitations

- Cross-sectional snapshot with no transaction timestamps
- Stated purchase frequency is not transaction-derived
- One observed purchase and one category per customer
- No acquisition cost, discount depth, or margin data

Data needed next

- Transaction timestamps for every order
- Promotion type and discount amount/depth
- Gross margin and fulfillment/return cost
- Acquisition source, acquisition date, and order-level category

Governing principle: *all rollout decisions require controlled experimentation and longitudinal tracking before being treated as proven.*